

GATE MOCK PAPER 2026

Data Science & Artificial Intelligence (DA)

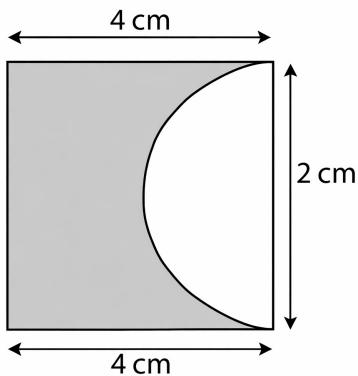
General Aptitude

Q.1 – Q.5 Carry ONE mark Each

Q.1	If “$\rightarrow$” denotes increasing order of magnitude (intensity), then Medium $\rightarrow$ Abundance $\rightarrow$ Plethora is analogous to: Scintilla $\rightarrow$ _____ $\rightarrow$ Inferno
(A)	Flame
(B)	Spark
(C)	Conflagration
(D)	Heat

Q.2	Read the following passage carefully. Some technological advances initially reduce employment in certain sectors. However, over time, they often create new types of jobs and industries that did not exist before. Which of the following must be TRUE based on the passage?
(A)	Technological advances always increase employment
(B)	Technological advances permanently eliminate jobs
(C)	Technological advances may initially reduce employment but can later generate new jobs
(D)	New industries prevent job losses caused by technology

Q.3	A fair six-sided die is thrown twice. What is the probability that the sum of the outcomes is a multiple of 3?
(A)	$\frac{1}{3}$
(B)	$\frac{5}{18}$
(C)	$\frac{1}{2}$
(D)	$\frac{7}{18}$

Q.4	The figure below shows a square of side 4 cm and a semicircle of radius 2 cm drawn inside it. The area (in cm^2) of the shaded region outside the semicircle but inside the square is: 
(A)	$16 - 2\pi$
(B)	$16 - \pi$
(C)	8π
(D)	4π

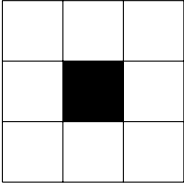
Q.5	The average of five numbers is 20. If one number 40 is replaced by 10, the new average is:
(A)	16
(B)	18
(C)	20
(D)	22

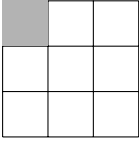
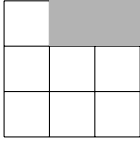
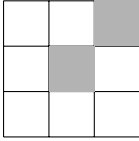
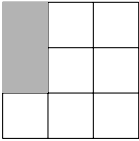
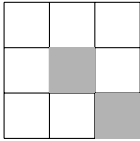
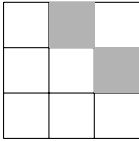
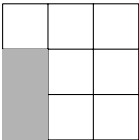
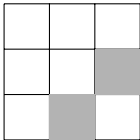
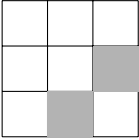
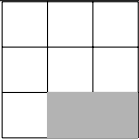
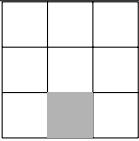
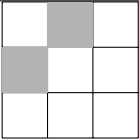
Q.6 – Q.10 Carry TWO marks Each

Q.6	Let $f(x) = x^2 - 4x + 3$. Which of the following statements are correct?
(A)	$f(x)$ has a minimum value
(B)	The minimum occurs at $x = 2$
(C)	$f(x) = 0$ has two real roots
(D)	The graph of $f(x)$ opens downward

Q.7	Two events A and B satisfy $P(A) = 0.6, P(B) = 0.5, P(A \cap B) = 0.3$ Which of the following are true?
(A)	A and B are independent
(B)	$P(A B) = 0.6$
(C)	$P(A \cup B) = 0.8$
(D)	$P(B A) = 0.5$

Q.8	Consider the infinite series $S = \sum_{n=1}^{\infty} (\sqrt{n+1} - \sqrt{n}).$ Which of the following statements is/are correct?
(A)	The series converges
(B)	The series diverges
(C)	The partial sum $S_n = \sqrt{n+1} - 1$
(D)	The series grows without bound but at a decreasing rate

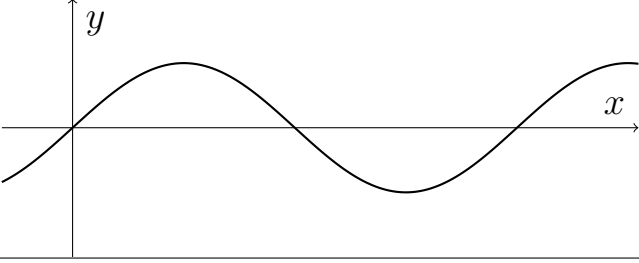
Q.9	A 3×3 grid is shown below. Initially, only the central square is shaded.  A square is shaded if and only if the number of its edge-adjacent shaded squares is even. The rule is applied simultaneously until no change occurs.
(A)	1
(B)	5
(C)	9
(D)	0

Q.10	Which one of the options completes the pattern?         ? (A) 
(B)	
(C)	
(D)	

Q.11 – Q.35 Carry ONE mark Each

Q.11	Consider the following statements: (i) If two random variables are independent, then they are uncorrelated. (ii) If two random variables are uncorrelated, then they are independent. Which one of the following options is correct?
(A)	Both (i) and (ii) are true.
(B)	Both (i) and (ii) are false.
(C)	(ii) is true and (i) is false.
(D)	(i) is true and (ii) is false.

Q.12	Three fair coins are tossed independently. Define random variables: X = number of heads, Y = number of tails. Let A be the event $X \geq 2$ and B be the event $Y \geq 2$. What is $P(A \cap B)$?
(A)	0
(B)	$\frac{1}{8}$
(C)	$\frac{3}{8}$
(D)	$\frac{1}{2}$

Q.13	Find the determinant of the matrix whose characteristic equation is represented by the graph below. Also find the rank of the matrix. 
(A)	$\det(A) = 0, \text{rank}(A) = 3$
(B)	$\det(A) = 1, \text{rank}(A) = 4$
(C)	$\det(A) \neq 0, \text{rank}(A) = 4$
(D)	$\det(A) = 0, \text{rank}(A) = 4$

Q.14	Consider the Python code: <pre>def f(x, L=[]): L.append(x) return L print(f(1)) print(f(2)) print(f(3, [])) print(f(4))</pre> What is the output?
(A)	[1], [2], [3], [4]
(B)	[1], [1,2], [3], [1,2,4]
(C)	[1], [2], [3], [3,4]
(D)	[1], [1,2], [3], [1,2,3,4]

Q.15	Consider the quadratic form $f(x, y, z) = 3x^2 + 3y^2 + 3z^2 - 2xy - 2xz - 2yz.$ Which of the following is correct?
(A)	Positive semi-definite
(B)	Negative definite
(C)	Positive definite
(D)	Neither positive nor negative definite

Q.16	Which statement about tail recursion is TRUE?
(A)	Always faster than iteration
(B)	Can always be optimized to $O(1)$ space
(C)	Uses implicit stack memory
(D)	May be optimized to iteration by compiler

Q.17	A vector in $\mathbb{R}^3$ is projected onto a subspace spanned by orthonormal vectors u and v to give w_1 . Now u and v are rotated by 180° , and the projection gives w_2 . Find $\ w_1 - w_2\ $.
(A)	0
(B)	$\ w_1\ $
(C)	$2\ w_1\ $
(D)	$\ v\ $

Q.18	What is true regarding the perceptron learning algorithm?
(A)	The algorithm always converges on any dataset
(B)	Smaller γ (margin), means there will be larger number of steps
(C)	The algorithm requires at least $\frac{R^2}{\gamma^2}$ iterations to converge
(D)	$f(x_n; x_{\text{new}}, b_{\text{new}}) > f(x_n; x_{\text{old}}, b_{\text{old}})$ for misclassified points

Q.19	Consider the two datasets A and B shown below. A • • • • × × × × B • • × × • • × × Which of the following statements is correct?
(A)	Single linkage correctly classifies B
(B)	Complete linkage correctly classifies A
(C)	Both clustering methods work fine
(D)	None of the above

Q.20	In which of the techniques, a Gaussian prior probability is assumed while building a linear regression model?
(A)	Ridge
(B)	Lasso
(C)	Both
(D)	None

Q.21	What is the worst-case time complexity of the following code?
	<pre> void foo(int n) { if (n <= 1) return; for (int i = 1; i <= n; i++) foo(n/2); } </pre>
(A)	$O(n \log n)$
(B)	$O(n^2)$
(C)	$O(n^n)$
(D)	$O(2^n)$

Q.22	Why is k -NN not preferred in high-dimensional datasets?
(A)	Distance becomes meaningless in high dimensions
(B)	The algorithm tends to overfit
(C)	Computationally expensive
(D)	All of the above

Q.23	Consider propositional logic with symbols p and q . Which one of the following formulas is a tautology?
(A)	$(p \wedge q) \rightarrow p$
(B)	$p \rightarrow (p \wedge q)$
(C)	$((p \rightarrow q) \wedge p) \rightarrow q$
(D)	$p \wedge \neg p$

Q.24	The joint PDF of random variables X and Y is given by $f_{X,Y}(x, y) = \begin{cases} 6y(1 - x), & 0 \leq y \leq x \leq 1, \\ 0, & \text{otherwise.} \end{cases}$ Find $E[Y \mid X = 0.6]$.

Q.25	An autonomous delivery robot uses the A* search algorithm with heuristic function $h(n)$. Which properties of the heuristic guarantee optimality of A*?
(A)	$h(n) \leq h^*(n)$ for all nodes
(B)	$h(n)$ is monotonic (consistent)
(C)	$h(n)$ may overestimate occasionally
(D)	$f(n) = g(n) + h(n)$ never decreases along a path

Q.26	A table T has 10 million rows. A B ⁺ -tree index exists on attribute X . Which query benefits least from the index?
(A)	SELECT * FROM T WHERE $X = 10$;
(B)	SELECT * FROM T WHERE X BETWEEN 10 AND 100;
(C)	SELECT * FROM T ORDER BY X ;
(D)	SELECT AVG(X) FROM T;

Q.27	Let Z be a standard normal random variable and $X = aZ + b$. Given $E[X] = 1$, $E[(X - E[X])Z] = -2$, $E[(X - E[X])^2] = 4$, find the values of a and b.
(A)	$a = -2, b = 1$
(B)	$a = 2, b = -1$
(C)	$a = -2, b = -1$
(D)	$a = 1, b = 1$

Q.28	Which traversal uniquely identifies a binary search tree?
(A)	Preorder
(B)	Inorder
(C)	Postorder
(D)	Level order

Q.29	A factory produces high quality microprocessors. On average, 0.2% are defective. Processors are packed in boxes of 1500 units. Using Poisson approximation, find the probability that a randomly selected box contains exactly 4 defective processors. $[e^{-3} \approx 0.0498]$
(A)	0.050
(B)	0.112
(C)	0.168
(D)	0.224

Q.30	A company stores employee records in a hash table of size 50 using linear probing. Currently, 35 slots are occupied. What is the maximum number of probes required in the worst case to insert a new employee record? _____

Q.31

An esports league maintains the following two tables **Player** and **Roster**. The attribute **id** in table **Roster** references the primary key of table **Player**.

Player Table

id	name	played
1	Alex	180
2	Boris	210
3	Chen	160
4	Diego	200
5	Ethan	195
6	Farhan	205

Roster Table

city	id	kills	assist
Tokyo	2	230	180
Seoul	3	190	175
Berlin	5	210	160
Tokyo	1	195	200
Seoul	4	205	190
Tokyo	6	215	185

SQL Query:

SELECT * FROM Player P, Roster R

WHERE P.id = R.id

AND city = 'Tokyo'

AND kills >= (SELECT AVG(kills) FROM Roster);

What is the number of^{d7} rows returned by the above SQL query?

Q.32	A complete binary tree has 31 nodes. What is its height? _____

Q.33

Three hospitals conduct a medical test as follows:

Hospital	P(patient goes)	Test accuracy
H1	40%	85%
H2	35%	90%
H3	25%	80%

A patient is randomly selected and tests positive. What is the probability that the test was conducted at hospital H2? (Round off to two decimal places.)

Q.34	Let A be a 4×4 matrix with eigenvalues $2, 2, -1, 3$. Find the value of $\det(A^2)$. _____

Q.35	Let A be a square matrix satisfying $A^2 = 8A - I.$ Find the value of $\det(A)$. _____

Q.36 – Q.65 Carry TWO marks Each

Q.36	Let $X_1, \dots, X_{150}$ be i.i.d. Poisson(2). Let $Y = \sum X_i$. Using CLT, approximate $P(260 \leq Y \leq 340)$.
(A)	$\Phi(1) - \Phi(-1)$
(B)	$\Phi(2) - \Phi(-2)$
(C)	$\Phi(3) - \Phi(-3)$
(D)	$\Phi(0.5) - \Phi(-0.5)$

Q.37	A website records whether a visitor clicks an ad. Each visitor clicks with probability p, independently. Let $\hat{p} = \frac{1}{n} \sum_{i=1}^n X_i,$ where $X_i = 1$ if the i-th visitor clicks. Which of the following statements is/are correct?
(A)	$\hat{p}$ is an unbiased estimator of p
(B)	$\text{Var}(\hat{p}) = p(1 - p)$
(C)	$\lim_{n \rightarrow \infty} \hat{p} = p$ almost surely
(D)	$\text{Var}(\hat{p})$ increases with n

Q.38	A hospital uses an AI system based on a Bayesian Network to diagnose a disease D using two symptoms S_1 (fever) and S_2 (fatigue). The structure of the Bayesian Network is: $D \rightarrow S_1, \quad D \rightarrow S_2$ Which of the following statements are correct?
(A)	S_1 and S_2 are conditionally independent given D
(B)	S_1 and S_2 are marginally independent
(C)	Observing S_1 changes the probability of S_2
(D)	Observing both S_1 and S_2 gives more information about D than observing only one

Q.39	Consider a Bayesian network with structure: $A \rightarrow B \rightarrow C \leftarrow D$ Which of the following statements are TRUE?
(A)	A is independent of D
(B)	A is independent of C given B
(C)	B is independent of D given C
(D)	Observing C makes A and D dependent

Q.40	When a matrix A is decomposed using Singular Value Decomposition, $A = U\Sigma V^T,$ with singular values $\sigma_1 = 7, \sigma_2 = 5, \sigma_3 = 4, \sigma_4 = 0$, which statements are correct?
(A)	$\text{rank}(A) = 2$
(B)	$\text{nullity}(A) = 1$
(C)	Column space of A is spanned by first three columns of U
(D)	Null space of A is spanned by last column of V

Q.41	For the function $f(x) = 3x^4 - 16x^3 + 30x^2 - 24x + 3,$ which of the following statements is/are correct?
(A)	$x = 2$ is a local minimum
(B)	$x = 2$ is a local maximum
(C)	$x = 5$ is a point of inflection
(D)	$x = 1$ is a local maximum

Q.42	Which of the following statements are TRUE?
(A)	Every candidate key is minimal
(B)	A superkey may have redundant attributes
(C)	Every relation has exactly one candidate key
(D)	Primary key must be chosen from candidate keys

Q.43	Consider: <pre>SELECT * FROM R WHERE A NOT IN (SELECT B FROM S);</pre> Which case makes the query return zero rows even if logical matches exist?
(A)	<i>A</i> contains NULL
(B)	<i>B</i> contains NULL
(C)	Both <i>A</i> and <i>B</i> contain NULL
(D)	<i>R</i> is empty

Q.44	Which of the following SQL queries are valid and meaningful?
(A)	SELECT dept, COUNT(*) FROM Employee GROUP BY dept;
(B)	SELECT dept, salary FROM Employee GROUP BY dept;
(C)	SELECT COUNT(DISTINCT dept) FROM Employee;
(D)	SELECT dept FROM Employee HAVING COUNT(*) > 3;

45	Which problems can be solved using greedy strategy?
(A)	Activity selection
(B)	Huffman coding
(C)	Fractional knapsack
(D)	0/1 knapsack

46	Which problems are NP-complete? (logical multi-correct classification)
(A)	Hamiltonian cycle
(B)	Bipartite checking
(C)	Minimum spanning tree
(D)	Shortest path

Q.47	Which of the following statements are TRUE about Bellman–Ford?
(A)	Works with negative edges
(B)	Detects negative cycles
(C)	Fails if negative edge exists
(D)	Always faster than Dijkstra

Q.48	A relation has 7 attributes and two candidate keys, each of size 2. How many prime attributes are there? _____

Q.49	A city has 9 locations connected as a tree. Minimum new roads required so that failure of any one road does not disconnect the network? _____

Q.50	Let X, Y be independent random variables. Which are correct?
(A)	$Cov(X, Y) = 0$
(B)	$E[XY] = E[X]E[Y]$
(C)	They must be normally distributed
(D)	They are uncorrelated

Q.51	Let A be an $n \times n$ real matrix such that $A^T A = I$ and $\det(A) = -1$. Which of the following statements are correct?
(A)	$A^{-1} = A^T$
(B)	A represents a reflection
(C)	All eigenvalues of A have magnitude 1
(D)	A must be symmetric

Q.52	Which statements are TRUE?
(A)	BCNF always implies 3NF.
(B)	3NF always implies BCNF.
(C)	BCNF decomposition may lose dependencies.
(D)	3NF guarantees lossless and dependency-preserving decomposition.

Q.53	Which statements about B ⁺ -tree indexes are TRUE?
(A)	All keys appear at the leaf level.
(B)	Leaf nodes are linked sequentially.
(C)	Height is independent of number of records.
(D)	Range queries are efficient.

Q.54	Consider the directed graph shown with negative edge weights but no negative cycle. Which of the following statements are TRUE?
(A)	Dijkstra's algorithm will fail on this graph
(B)	Bellman–Ford algorithm can correctly compute shortest paths
(C)	Presence of a negative edge implies a negative cycle
(D)	Bellman–Ford can detect negative cycles if they exist

Q.55	A standard deck of 52 cards containing 4 aces and 12 face cards (Jack, Queen, King) is thoroughly shuffled and dealt equally among 4 players, such that each player receives 13 cards. What is the probability that each player receives exactly one ace and exactly three face cards?
(A)	$\frac{4! (12!) (36!) (13!)^2}{(3!)^4 (9!)^4 (52!)}$
(B)	$\frac{4^4 \binom{12}{3}^4 \binom{36}{9}^4}{\binom{52}{13}^4}$
(C)	$\frac{\binom{4}{1}^4 \binom{12}{3}^4 \binom{36}{9}^4}{(52!)/(13!)^4}$
(D)	$\frac{4! \binom{12}{3} \binom{36}{9}}{\binom{52}{13}}$

Q.56	Let $\{X_n\}_{n \geq 1}$ be a sequence of random variables such that for any $n \geq 1$ the following holds: $P(X_n = 0) = 1 - n^{-1}, \quad P(X_n = 1) = n^{-1}.$ Which of the following is/are true?
(A)	X_n converges to 0 in mean
(B)	X_n converges to 0 in distribution
(C)	X_n converges to 0 almost surely
(D)	X_n converges to 0 in mean squared sense

Q.57	What are the advantages of using Gini impurity over information gain?
(A)	Gini impurity is more sensitive to changes in class probability
(B)	Gini impurity is computationally more efficient
(C)	Gini impurity produces more balanced trees
(D)	Gini impurity performs better on highly imbalanced data

Q.58	Let A be a 3×3 matrix with eigenvalues 2, 3, 5. Find the trace of A^{-1} . _____

Q.59	In a two-player zero-sum game, alpha-beta pruning is applied to the minimax search tree. Which of the following statements is TRUE?
(A)	Alpha-beta pruning always reduces the time complexity to $O(b^{d/2})$
(B)	Alpha-beta pruning changes the final minimax value
(C)	Alpha-beta pruning prunes nodes that cannot affect the final decision
(D)	Alpha-beta pruning works only for deterministic games

Q.60	Let $f(x) = \begin{cases} \frac{\sin(ax)}{x}, & x \neq 0, \\ b, & x = 0, \end{cases}$ where a and b are real constants. Which of the following statements are correct?
(A)	$f(x)$ is continuous at $x = 0$ only if $b = a$
(B)	$f(x)$ is differentiable at $x = 0$ only if $a = 0$
(C)	If $a \neq 0$, then $f(x)$ is not differentiable at $x = 0$
(D)	If $a = 2$, then the derivative at $x = 0$ equals 0

Q.61	A min-priority queue is implemented using a binary heap. Initially the heap contains $n = 1023$ elements. A sequence of operations is performed: (i) Insert 1 element, (ii) Perform Extract-Min once. What is the maximum possible number of key comparisons performed during these two operations combined? _____

Q.62	A binary min-heap initially contains 15 distinct elements. A sequence of operations is performed: Insert 5 new elements, then perform 4 Extract-Min operations. What is the final number of nodes in the heap? _____

Q.63	A full binary game tree of depth 4 is searched using the minimax algorithm without pruning. How many leaf nodes are evaluated? _____

Q.64	A relation $R(A, B, C, D, E)$ has functional dependencies: $A \rightarrow BC$, $CD \rightarrow E$, $B \rightarrow D$. What is the total number of candidate keys of R? _____

Q.65	A B⁺-tree index has order 4 (max 4 children per internal node). What is the minimum possible height of the tree needed to index 2000 keys? _____